6 Random Variable and Mathematical Expectation



Simeon-Denis Poisson
(June 21, 1781 – April 25, 1840)

Introduction

Between Sylvestre-François Lacroix in 1816 and Louis Bachelier in 1914, the concepts random variable and mean of a random variable were invented. Some authors contend that Simeon-Denis Poisson invented the concepts random variable and expected value. Simeon-Denis Poisson, a French mathematician known for his work on probability theory. Poisson's research was mainly on Probability. In 1838, Poisson published his ideas on probability theory, which also included what is now known as Poisson distribution. He published between 300 and 400 mathematical works including applications to electricity and magnetism, and astronomy.

Learning Objectives

After studying this chapter students are able to understand

- Why do we use random variable?
- Why do we need to define the random variable?
- the types of Random variables.
- the probability function.
- the distribution function.
- the nature of problems.
- the methods for studying random experiments with outcomes that can be described numerically.
- the real probabilistic computation.
- the concept of mathematical expectation for discrete and continuous random variables.
- the properties of mathematical expectation for discrete and continuous random variables.
- the determination of mean and variance for discrete and continuous random variables.
- and the practical applications of mathematical expectations for discrete and continuous random variables.

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